QP CODE: 24900082

Reg No:.....

Name:.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

First Semester

Discipline Specific Core Course - MG1DSCECT101 - COMPUTER FUNDAMENTALS AND BASICS OF PC HARDWARE

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain][CO No(s)]

Part A

Multiple Choice Questions

Answer all questions.Each question carries 1 mark

1 Identify the smallest unit of data in computing [U] [1]

a) Bit b) Byte

c) Kilobyte d) Megabyte

2 Identify the component of CPU responsible for executing instructions. [U] [1]

a) Control Unit b) Arithmetic Logic Unit (ALU)

c) Register d) Cache

3 Identify which one is not a primary function of an operating system. [U] [1]

a) Memory management b) Process management

c) Application development d) File management

4 Define data. [K] [1]

a) Raw facts and figures
b) A type of software
c) A form of hardware
d) Processed information

5 Select the primary function of the Control Unit (CU) in a CPU [U] [2]

a) To perform mathematical calculations
b) To manage the execution of instructions and coordinate activities of the CPU
c) To store data temporarily
d) To execute input/output operations

6 Identify a type of primary memory among the following [U] [2]

a) Floppy Disk
b) Random Access Memory (RAM)
c) External USB Drive
d) Hard Disk Drive (HDD)

7 Select the correct purpose of the HDMI (High-Definition Multimedia Interface) port? [U] [2]

a) To connect a keyboard
b) To connect a mouse
c) To connect a monitor or Projectorfor video output
d) To connect a power cord

8 Examine which practice best supports green computing principles: [An] [3]

a) Disposing old computers
b) Virtualizing data storage
c) Frequent hardware upgrades
d) Using CRT monitors

9 Compare the following fields and determine which one benefits the most from neuromorphic chip technology [An] [3]

a) Artificial intelligence
b) Digital photography
c) Quantum mechanics
d) Traditional software programming

10 Differentiate between L1 and L3 cache. [An] [3]

a) L1 is faster and smaller than L3
b) L3 is faster and smaller than L1
c) Both are the same size
d) L1 is slower than L3

(10 × 1 = 10)

Part B

Short Answer Questions

Answer 4 questions. Each question carries 5 marks

11	Describe the step-by-step process of how a computer processes data, from input to output.	[U]	[1]
12	Differentiate between primary memory and secondary memory. Give examples of each.	[U]	[1]
13	Explain the key features and functions of flash drives.	[U]	[2]
14	Classify various types of scanners? Provide a brief explanation of at least two types.	[U]	[2]
15	Show how latency play in memory technology, and how is it managed in DDR4 and DDR5?	[An]	[3]
16	Distinguish AR from VR in terms of user experience.	[An]	[3]

(4 × 5 = 20)

Part C

Essay Questions

Answer 2 questions. Each question carries 10 marks

17	Explain the characteristics of fifth generation computers.	[U]	[1]
18	Define software. Differentiate between system software and application software, providing examples of each.	[U]	[1]
19	Compare and Contrast the functionality and limitation of PCI and PCIe exansion slots.	[U]	[2]
20	Distinguish between the different levels of cache in a CPU and their impact on multi-core processor performance.	[An]	[3]

(2 × 10 = 20)

END OF THE QUESTION PAPER

***